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Digital RF-Receiving Technology and Electron Paramagnetic Resonance

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Abstract

This experiment investigated both radio-frequency signal processing and continuous-wave electron paramagnetic resonance (EPR) spectroscopy using software-defined radio techniques. Frequency mixing was demonstrated with a laboratory RF mixer by observing the expected sum and difference frequencies and their evolution as one input frequency was varied. Radio signals in the VLF, VHF and UHF ranges were received and analysed using the RSP1 receiver and HDSDR software. Continuous-wave transmissions, FM broadcast signals and PMR transmissions were successfully identified from their spectral and temporal characteristics.

In the EPR measurements, dispersion curves of DPPH and ultramarine samples were recorded using a modified low-noise block converter (LNB). The resonance trajectories were extracted from HDSDR waterfall images by image processing and fitted with an empirical reciprocal model to determine the resonance field and microwave frequency. Three independent DPPH measurements yielded resonance fields between 0.356 T and 0.360 T with calculated Landé factors between $g = 2.094$ and 2.113. Measurements of ultramarine samples with different dilution ratios demonstrated that the EPR signal remained detectable over a wide concentration range, although increasing dilution resulted in larger uncertainties in the extracted resonance parameters. Comparison of the direct-mixing receiver with a calibrated PLL-LNB showed that the reconstructed microwave frequencies differed by only approximately 0.013 %. The dominant source of uncertainty in the EPR measurements was found to be the reconstruction and calibration of the magnetic-field axis rather than the microwave-frequency determination.

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1. Introduction

Electron paramagnetic resonance (EPR), also known as electron spin resonance, is a spectroscopic technique used to investigate atoms, molecules, and solids containing unpaired electrons. Such systems are paramagnetic and possess a magnetic moment associated with the electron spin. When a static external magnetic field B_0 is applied, the degeneracy of the electron-spin states is lifted by the electronic Zeeman interaction. For an effective spin-$\frac{1}{2}$ system, the energy separation between the two spin states is

$$\Delta E = g\mu_{\mathrm{B}} B_0, \tag{1}$$

where g is the Landé factor and μ_{B} is the Bohr magneton. Resonant transitions occur when the energy of the applied microwave radiation matches this splitting,

$$h\nu = g\mu_{\mathrm{B}} B_0. \tag{2}$$

The resonance field and microwave frequency can therefore be used to determine the g-factor, which provides information about the electronic state and local environment of the paramagnetic centre [**epr_manual**].

The experiment combined the physical principles of EPR with practical techniques used in radio-frequency receiving systems. In the first part, fundamental processes such as frequency mixing, amplitude modulation, frequency modulation, heterodyne reception, and software-defined radio were investigated. A frequency mixer converts two input signals into components at their sum and difference frequencies. The same principle is used in radio receivers to translate a high-frequency signal to a lower intermediate frequency that can be filtered, digitized, and analysed more conveniently. Using the RSP1 software-defined radio receiver and the HDSDR software, signals were studied both as frequency spectra and as time-dependent waterfall diagrams. Selected signals in the VLF, VHF, and UHF ranges were recorded in order to examine their carrier frequencies, bandwidths, modulation characteristics, and temporal structures.

In the EPR part of the experiment, a modified low-noise block downconverter was used as a microwave excitation and receiving system. Two local oscillators operating near 9.75 GHz and 10.6 GHz were involved in the direct-mixing arrangement. The sample was positioned inside the dielectric resonator and exposed to the microwave field, while an electromagnet produced the static magnetic field. A predefined magnetic-field sequence swept the field through the resonance condition. The high-frequency response was subsequently converted to an intermediate-frequency range near 800 MHz, where the characteristic EPR dispersion curves could be recorded using HDSDR.

DPPH was used as a reference material because it is a stable organic radical with a narrow EPR signal and a well-known g-factor close to that of a free electron. Measurements with DPPH samples of different masses were used to determine the resonance position and to investigate the dependence of the observable signal on the number of paramagnetic centres. Ultramarine samples were measured in the pure state and at dilution ratios of 1:10, 1:100, and 1:1000 in order to study how dilution affects the detectability and shape of the EPR response. Finally, the 10 µg DPPH sample was measured again using a calibrated PLL-LNB with a precisely specified local-oscillator frequency of 9.75 GHz. This allowed the direct-mixing and calibrated observation systems to be compared with respect to the reconstructed microwave frequency and the accuracy of the resonance determination.

The main objectives of the experiment were therefore to verify sum- and difference-frequency generation in an RF mixer, examine modulation and software-defined radio reception over several frequency ranges, observe EPR dispersion curves from DPPH and ultramarine, determine resonance frequencies, magnetic fields, and g-factors from the measured data, investigate the practical sensitivity of the setup, and evaluate the advantages of a calibrated local oscillator for quantitative EPR measurements. The measurements followed the revised six-task experimental programme, while the older EPR manual was used primarily as a source for the underlying resonance theory [**epr_tasks**, **epr_manual**].

2. Theoretical Background

2.1 Sinusoidal signals and spectral representation

A monochromatic electrical signal can be represented by a sinusoidal voltage

$$u(t) = U_0 \cos(2\pi f t + \phi)\,, \tag{3}$$

where U_0 is the peak voltage, f is the frequency and ϕ is the phase. For a sinusoidal voltage, the root-mean-square voltage is related to the peak and peak-to-peak voltages by

$$U_{\mathrm{rms}} = \frac{U_0}{\sqrt{2}} = \frac{U_{\mathrm{pp}}}{2\sqrt{2}}. \tag{4}$$

The mean power delivered to a matched resistance R is

$$P = \frac{U_{\mathrm{rms}}^2}{R}. \tag{5}$$

Equation (5) is applicable when the quoted voltage is the voltage across the load. If the generator amplitude is specified for a different termination condition, the corresponding impedance correction must be considered.

An oscilloscope displays the signal in the time domain, whereas a spectrum analyser or software-defined radio displays the relative contributions of different frequencies. The continuous Fourier transform of a time-dependent signal is

$$\widetilde{u}(f) = \int_{-\infty}^{\infty} u(t)\, \exp(-i2\pi f t)\, \mathrm{d}t. \tag{6}$$

For an ideal infinitely long sinusoid, the frequency spectrum consists of a single component at its oscillation frequency. Finite observation time, modulation, noise and non-linear components broaden the measured spectrum or produce additional spectral lines.

2.2 Frequency mixing

Frequency mixing is a non-linear process in which two input signals are combined to produce components at new frequencies. For two sinusoidal input voltages,

$$u_{\mathrm{L}}(t) = U_{\mathrm{L}} \cos(2\pi f_{\mathrm{L}} t + \phi_{\mathrm{L}}) \,, \tag{7}$$

and

$$u_{\mathrm{R}}(t) = U_{\mathrm{R}} \cos(2\pi f_{\mathrm{R}} t + \phi_{\mathrm{R}}) \,, \tag{8}$$

an ideal mixer produces an output proportional to their product,

$$u_{\mathrm{X}}(t) \propto u_{\mathrm{L}}(t) u_{\mathrm{R}}(t). \tag{9}$$

Using the trigonometric identity

$$\cos\alpha \cos\beta = \frac{1}{2} \left[\cos(\alpha+\beta) + \cos(\alpha-\beta)\right], \tag{10}$$

the output becomes

$$u_{\mathrm{X}}(t) \propto \frac{U_{\mathrm{L}} U_{\mathrm{R}}}{2} \Big[\cos(2\pi(f_{\mathrm{L}} + f_{\mathrm{R}})t + \phi_{\mathrm{L}} + \phi_{\mathrm{R}}) + \cos(2\pi(f_{\mathrm{L}} - f_{\mathrm{R}})t + \phi_{\mathrm{L}} - \phi_{\mathrm{R}}) \Big]. \tag{11}$$

The two principal mixing products are therefore

$$f_{+} = f_{\mathrm{L}} + f_{\mathrm{R}}, \qquad f_{-} = |f_{\mathrm{L}} - f_{\mathrm{R}}| \,. \tag{12}$$

In a real mixer, the original input frequencies may also appear at the output because of imperfect isolation between the ports. Harmonics and intermodulation products of the general form

$$f_{n,m} = |n f_{\mathrm{L}} \pm m f_{\mathrm{R}}| \,, \qquad n, m \in \mathbb{N}, \tag{13}$$

may also be present. Their amplitudes depend on the mixer linearity, input powers, conversion loss, port isolation, impedance matching and whether the mixer is overdriven.

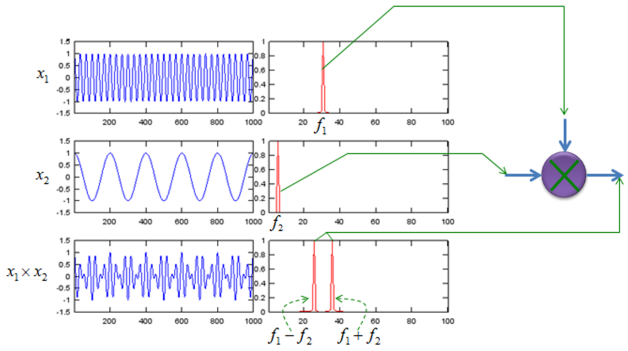


Figure 1: Ideal frequency mixing in the time and frequency domains. Two sinusoidal input signals produce components at the sum and difference frequencies. Adapted from [1].

2.3 Amplitude and frequency modulation

Modulation transfers information from a comparatively low-frequency message signal to a higher-frequency carrier. In amplitude modulation, the amplitude of the carrier varies with the message signal. For a sinusoidal message, the modulated voltage is

$$u_{\mathrm{AM}}(t) = U_{\mathrm{c}} \left[1 + m \cos(2\pi f_{\mathrm{m}} t)\right] \cos(2\pi f_{\mathrm{c}} t), \tag{14}$$

where f_{c} is the carrier frequency, f_{m} is the modulation frequency and m is the modulation depth. Expanding Eq. (14) gives

$$\begin{aligned} u_{\mathrm{AM}}(t) =& U_{\mathrm{c}} \cos(2\pi f_{\mathrm{c}} t) \\ &+ \frac{m U_{\mathrm{c}}}{2} \cos[2\pi(f_{\mathrm{c}} + f_{\mathrm{m}})t] \\ &+ \frac{m U_{\mathrm{c}}}{2} \cos[2\pi(f_{\mathrm{c}} - f_{\mathrm{m}})t] \,. \end{aligned} \tag{15}$$

The AM spectrum therefore contains the carrier at f_{c} and two sidebands at

$$f_{\mathrm{USB}} = f_{\mathrm{c}} + f_{\mathrm{m}}, \qquad f_{\mathrm{LSB}} = f_{\mathrm{c}} - f_{\mathrm{m}}. \tag{16}$$

In frequency modulation, the instantaneous carrier frequency is varied. For sinusoidal modulation,

$$u_{\mathrm{FM}}(t) = U_{\mathrm{c}} \cos[2\pi f_{\mathrm{c}} t + \beta \sin(2\pi f_{\mathrm{m}} t)] \,, \tag{17}$$

where the modulation index is

$$\beta = \frac{\Delta f}{f_{\mathrm{m}}}, \tag{18}$$

and Δf is the peak frequency deviation. Unlike simple single-tone AM, an FM signal generally contains several pairs of sidebands separated by f_{m}. Their relative amplitudes depend on β.

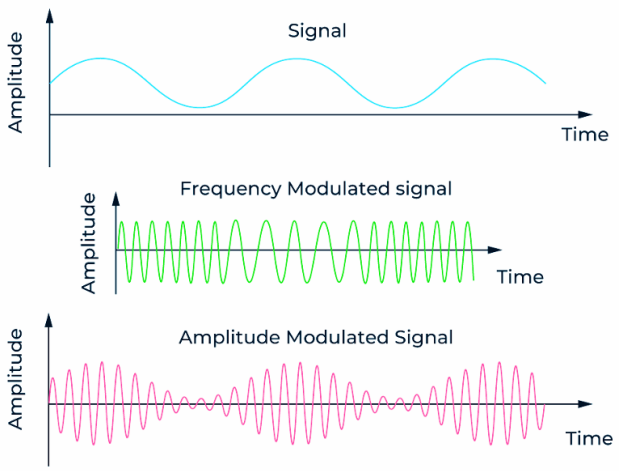


Figure 2: Comparison of amplitude and frequency modulation in the time domain. In amplitude modulation, the carrier envelope changes, whereas in frequency modulation the instantaneous carrier frequency changes. Adapted from [2].

2.4 Software-defined radio and heterodyne reception

A software-defined radio converts an analogue receiver signal into digital samples. Filtering, gain control, spectral analysis, demodulation and waterfall visualisation are subsequently carried out numerically. The waterfall display represents frequency along the horizontal axis and successive acquisition times along the vertical axis, while the colour or brightness indicates the measured signal strength.

Radio receivers commonly use heterodyne frequency conversion. An incoming radio-frequency signal f_{RF} is mixed with a local oscillator f_{LO}, producing an intermediate frequency

$$f_{\mathrm{IF}} = \left| f_{\mathrm{RF}} - f_{\mathrm{LO}} \right| . \tag{19}$$

The original radio frequency is consequently reconstructed using

$$f_{\mathrm{RF}} = f_{\mathrm{LO}} \pm f_{\mathrm{IF}}. \tag{20}$$

The sign in Eq. (20) cannot be determined from the intermediate frequency alone. The appropriate sideband must be selected from the known operating range and the receiver configuration. A stable and accurately known local oscillator is therefore essential when the absolute RF frequency is required.

The frequency resolution of a Fourier spectrum is limited principally by the effective acquisition time T_{obs}. As a first approximation,

$$\Delta f_{\mathrm{res}} \approx \frac{1}{T_{\mathrm{obs}}}. \tag{21}$$

In practice, the displayed resolution is also affected by the selected resolution bandwidth, sampling rate, spectral window, averaging and digital filtering.

2.5 Interpretation of VLF, VHF and UHF signals

A received radio signal can be characterised through its centre frequency, occupied bandwidth, signal-to-noise ratio and temporal behaviour. In a waterfall display, an unmodulated continuous-wave carrier ideally appears as a narrow line at a nearly constant frequency. An FM broadcast signal occupies a substantially broader frequency interval because the carrier frequency changes continuously with the transmitted audio signal.

Digitally coded or periodically transmitted signals may appear as bursts, repeated interruptions, discrete frequency shifts or time-dependent changes in intensity. The occupied bandwidth may be defined from lower and upper frequency limits as

$$\Delta f_{\mathrm{occ}} = f_{\mathrm{upper}} - f_{\mathrm{lower}}. \tag{22}$$

The measured signal-to-noise ratio may be expressed using linear amplitudes as

$$\mathrm{SNR} = \frac{A_{\mathrm{signal}}}{A_{\mathrm{noise}}}, \tag{23}$$

or in decibels as

$$\mathrm{SNR}_{\mathrm{dB}} = 20 \log_{10} \left(\frac{A_{\mathrm{signal}}}{A_{\mathrm{noise}}} \right). \tag{24}$$

Signal identification should not be based on frequency alone. The observed frequency, bandwidth, modulation pattern, repetition time and antenna dependence must be compared with the expected properties of the proposed transmitter.

2.6 Electron Zeeman interaction

Electron paramagnetic resonance is observed in systems containing unpaired electrons. For an effective electron spin $S = \frac{1}{2}$ in a static magnetic field B_0, the two allowed spin projections are

$$m_s = -\frac{1}{2}, \quad +\frac{1}{2}. \tag{25}$$

The Zeeman energy of a spin state is

$$E_{m_s} = g \mu_{\mathrm{B}} B_0 m_s, \tag{26}$$

where g is the Landé factor and μ_{B} is the Bohr magneton. The energy separation between the two spin states is

$$\Delta E = g \mu_{\mathrm{B}} B_0. \tag{27}$$

The splitting therefore increases linearly with the applied magnetic field. Microwave radiation can induce transitions between the two states when the magnetic-dipole selection rule

$$\Delta m_s = \pm 1 \tag{28}$$

is satisfied.

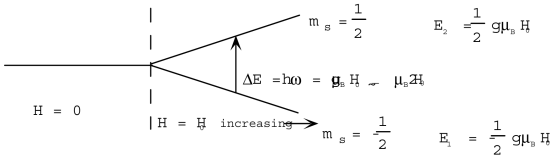


Figure 3: Schematic Zeeman splitting of a spin-$\frac{1}{2}$ system under an external static magnetic field. The original diagram was presented for a spin-$\frac{1}{2}$ nucleus; here, the analogous level splitting is used to illustrate the electron-spin states in EPR. Adapted from [3].

2.7 EPR resonance condition and the g-factor

A photon of frequency ν has the energy

$$E_\gamma = h\nu, \tag{29}$$

where h is Planck's constant. Resonance occurs when the photon energy equals the Zeeman splitting,

$$h\nu_{\mathrm{res}} = g\mu_{\mathrm{B}} B_{\mathrm{res}}. \tag{30}$$

The experimental g-factor can therefore be calculated from

$$g_{\mathrm{exp}} = \frac{h\nu_{\mathrm{res}}}{\mu_{\mathrm{B}} B_{\mathrm{res}}}. \tag{31}$$

Alternatively, when the g-factor and microwave frequency are known, the expected resonance field is

$$B_{\mathrm{res,expected}} = \frac{h\nu_{\mathrm{res}}}{g\mu_{\mathrm{B}}}. \tag{32}$$

The g-factor of a free electron is approximately 2.0023. Interactions with the chemical and solid-state environment can shift the effective g-factor away from the free-electron value. The measured g-factor therefore provides information about the electronic environment of the paramagnetic centre.

2.8 Static and oscillating magnetic fields

The static magnetic field B_0 defines the quantisation axis and produces the Zeeman splitting. The microwave field contains an oscillating magnetic component B_1 perpendicular to B_0. A simplified total field can be written as

$$\mathbf{B}(t) = B_0 \mathbf{e}_z + 2B_1 \cos(\omega t)\mathbf{e}_x. \tag{33}$$

A linearly oscillating field may be decomposed into two oppositely rotating circularly polarised components. Only the component rotating in the same sense as the electron-spin precession efficiently drives the transition.

In a conventional continuous-wave EPR experiment, the microwave frequency is held approximately constant while the static magnetic field is swept through resonance. For an approximately linear field sweep,

$$B(t) = B_{\mathrm{start}} + \frac{B_{\mathrm{end}} - B_{\mathrm{start}}}{T_{\mathrm{sweep}}} t. \tag{34}$$

The corresponding sweep rate is

$$\frac{\mathrm{d}B}{\mathrm{d}t} = \frac{B_{\mathrm{end}} - B_{\mathrm{start}}}{T_{\mathrm{sweep}}}. \tag{35}$$

Equations (34) and (35) are later used to convert the HDSDR waterfall time axis into a magnetic-field axis.

2.9 Absorption, dispersion and field-modulated detection

The response of a spin system to an oscillating magnetic field can be described by the complex magnetic susceptibility

$$\chi(\omega) = \chi'(\omega) - i\chi''(\omega). \tag{36}$$

The imaginary component χ'' describes microwave absorption, whereas the real component χ' describes the dispersive response. An ideal Lorentzian absorption line may be represented by

$$\chi''(B) \propto \frac{\Gamma_B}{(B - B_{\mathrm{res}})^2 + \Gamma_B^2}, \tag{37}$$

where Γ_B is the half width at half maximum. The associated dispersion line has the form

$$\chi'(B) \propto \frac{B - B_{\mathrm{res}}}{(B - B_{\mathrm{res}})^2 + \Gamma_B^2}. \tag{38}$$

The full width at half maximum of the absorption line is

$$\Delta B_{\mathrm{FWHM}} = 2\Gamma_B. \tag{39}$$

In conventional continuous-wave EPR, a small alternating modulation field is superimposed on the slowly swept static field,

$$B_{\mathrm{tot}}(t) = B_0(t) + B_{\mathrm{mod}} \cos(2\pi f_{\mathrm{mod}} t). \tag{40}$$

When the modulation amplitude is small compared with the linewidth, phase-sensitive detection produces a signal proportional to the first derivative of the absorption line,

$$S_{\mathrm{mod}}(B) \propto \frac{\mathrm{d}\chi''(B)}{\mathrm{d}B}. \tag{41}$$

At the centre of the absorption line, the derivative is zero. The largest positive and negative signals occur near the steepest sides of the absorption peak.

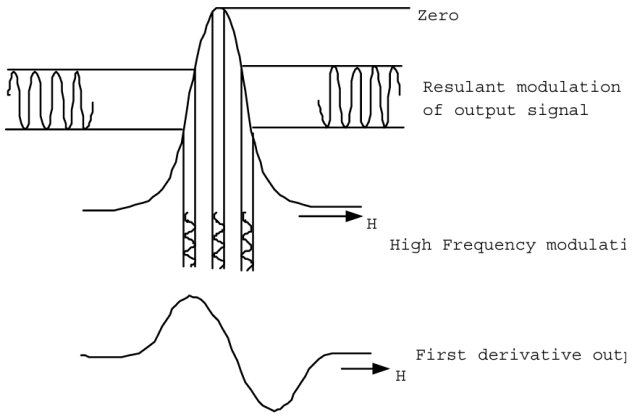


Figure 4: Relation between the EPR absorption spectrum and its first-derivative representation. Adapted from [4].

The HDSDR measurements in the present experiment did not directly record the conventional derivative amplitude shown in Fig. 4. Instead, HDSDR tracked the frequency-dependent response of the modified LNB while the magnetic field was swept. The observed S-shaped or asymptotic waterfall feature is therefore referred to as a dispersion curve, but should not be identified automatically with either $\chi'(B)$ or $\mathrm{d}\chi''/\mathrm{d}B$ without considering the complete receiver response.

2.10 Linewidth and relaxation

The width of an EPR line is influenced by intrinsic spin relaxation, spin–spin interactions, magnetic-field inhomogeneity, unresolved hyperfine interactions, sample concentration, microwave-power saturation and instrumental broadening.

The transverse relaxation time T_2 describes the loss of phase coherence within the spin ensemble. For an ideal Lorentzian line in the frequency domain,

$$T_2 \approx \frac{1}{\pi \Delta \nu_{\mathrm{FWHM}}}, \tag{42}$$

where $\Delta\nu_{\mathrm{FWHM}}$ is the frequency-domain full width at half maximum. Equation (42) is only applicable when the measured line is approximately Lorentzian and the observed width is not dominated by field inhomogeneity, sweep-rate effects or receiver instability.

The longitudinal relaxation time T_1 describes the return of the spin populations to thermal equilibrium through energy exchange with the environment. Excessive microwave power can equalise the populations of the spin states faster than relaxation restores the equilibrium difference, causing saturation and a reduction or distortion of the EPR signal.

2.11 DPPH and ultramarine samples

DPPH is a stable organic radical with a narrow EPR line and a well-established reference g-factor of approximately

$$g_{\mathrm{DPPH}} \approx 2.0037. \tag{43}$$

It is therefore suitable for checking the relationship between microwave frequency and resonance field and for evaluating the practical sensitivity of the setup.

In the unsaturated regime, and for otherwise unchanged measurement conditions, the measured EPR amplitude is expected to scale approximately with the number of contributing paramagnetic centres,

$$A_{\mathrm{EPR}} \propto N_{\mathrm{spin}}. \tag{44}$$

For samples of the same material and density, the number of spins is approximately proportional to the sample mass,

$$N_{\mathrm{spin}} \propto m_{\mathrm{sample}}. \tag{45}$$

Thus, under unchanged instrumental conditions,

$$A_{\mathrm{EPR}} \propto m_{\mathrm{sample}}. \tag{46}$$

Deviations from Eq. (46) may result from changes in receiver gain, sample placement, resonator filling, microwave saturation, field inhomogeneity or noise.

Ultramarine also contains paramagnetic centres that produce a detectable EPR response. Diluting the material reduces the number of active centres inside the resonator and should therefore primarily reduce the signal strength. The resonance position and intrinsic g-factor should remain approximately constant if the chemical identity and local environment of the paramagnetic centres are unchanged. Changes in packing, resonator loading, sample position and signal-to-noise ratio can nevertheless shift the experimentally determined centre or alter the apparent linewidth.

For a dilution ratio $1 : r$, the relative concentration is

$$c_{\mathrm{rel}} = \frac{1}{r}. \tag{47}$$

2.12 Modified LNB and calibrated PLL-LNB detection

A low-noise block downconverter combines low-noise amplification, local oscillators and frequency mixing. In the modified direct-mixing arrangement used for the EPR measurements, oscillator signals near

$$f_{\mathrm{LO},1} \approx 9.75\,\mathrm{GHz} \tag{48}$$

and

$$f_{\mathrm{LO},2} \approx 10.6\,\mathrm{GHz} \tag{49}$$

were involved. The sample was placed in the dielectric resonator and interacted with the microwave field, while the received response was converted to an intermediate-frequency range near 800 MHz, where it could be recorded

by the RSP1 receiver and HDSDR.

For a single local oscillator, the detected intermediate frequency is

$$f_{\mathrm{IF}} = \left| f_{\mathrm{sample}} - f_{\mathrm{LO}} \right|. \tag{50}$$

The physical microwave frequency is reconstructed using

$$f_{\mathrm{sample}} = f_{\mathrm{LO}} \pm f_{\mathrm{IF}}. \tag{51}$$

In the direct-mixing system, uncertainty in the oscillator frequencies, oscillator drift and the presence of more than one mixing path can limit the accuracy with which the absolute microwave frequency is reconstructed.

For the calibrated PLL-LNB used in Task 6, the local-oscillator frequency was specified as

$$f_{\mathrm{LO,PLL}} = 9.75\,\mathrm{GHz}. \tag{52}$$

For the upper mixing branch used in the later analysis,

$$f_{\mathrm{RF,PLL}} = f_{\mathrm{LO,PLL}} + f_{\mathrm{IF}}. \tag{53}$$

The PLL stabilises the receiver local oscillator and therefore reduces one contribution to the microwave-frequency uncertainty. It does not, however, remove uncertainties caused by the magnetic-field calibration, the sample position, resonator loading, excitation-source drift or the extraction of the resonance position from the waterfall image.

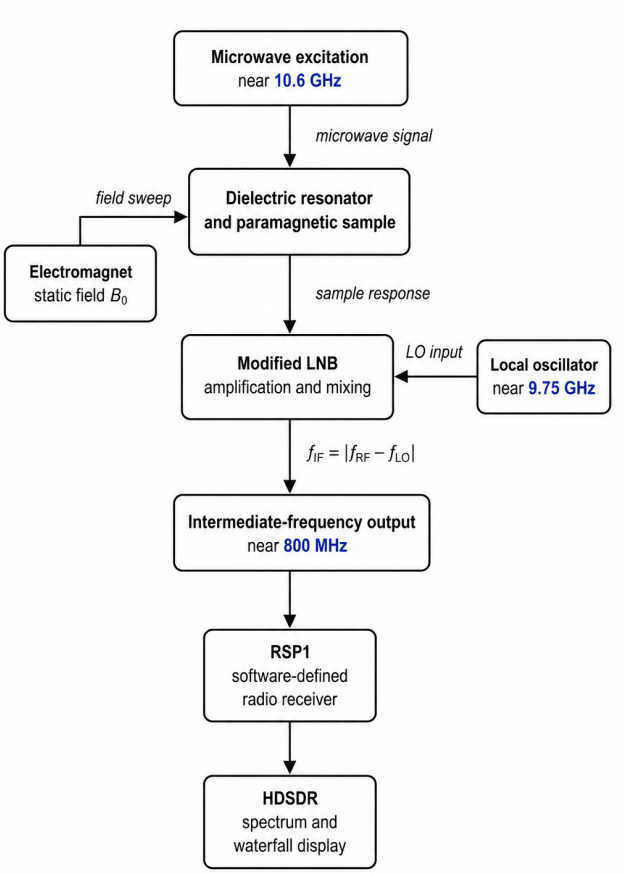


Figure 5: Simplified frequency-conversion chain of the modified LNB-based EPR detection system. The microwave response of the resonator–sample system is mixed with a local oscillator near 9.75 GHz, producing an intermediate-frequency signal near 800 MHz that is recorded using the RSP1 receiver and HDSDR.

2.13 Uncertainty propagation and comparison quantities

For a quantity calculated from independent measured variables, the standard uncertainty is obtained through Gaussian error propagation. For the g-factor in Eq. (31),

$$\frac{u(g)}{g} = \sqrt{\left(\frac{u(\nu_{\mathrm{res}})}{\nu_{\mathrm{res}}}\right)^2 + \left(\frac{u(B_{\mathrm{res}})}{B_{\mathrm{res}}}\right)^2}. \tag{54}$$

The absolute uncertainty is therefore

$$u(g) = g\sqrt{\left(\frac{u(\nu_{\mathrm{res}})}{\nu_{\mathrm{res}}}\right)^2 + \left(\frac{u(B_{\mathrm{res}})}{B_{\mathrm{res}}}\right)^2}. \tag{55}$$

For the reconstructed frequency in Eq. (51),

$$u(f_{\mathrm{RF}}) = \sqrt{u(f_{\mathrm{LO}})^2 + u(f_{\mathrm{IF}})^2}. \tag{56}$$

For the difference between two independent quantities,

$$\Delta x = x_2 - x_1, \tag{57}$$

the uncertainty is

$$u(\Delta x) = \sqrt{u(x_1)^2 + u(x_2)^2}. \tag{58}$$

The relative difference between two measured frequencies may be reported as

$$\delta_f = \frac{|f_2 - f_1|}{f_{\mathrm{ref}}}. \tag{59}$$

When expressed as a percentage,

$$\delta_{f,\%} = 100\,\delta_f. \tag{60}$$

The relative deviation of an experimental value from a reference value is similarly calculated using

$$\varepsilon_{\%} = 100\frac{\left|x_{\mathrm{exp}} - x_{\mathrm{ref}}\right|}{|x_{\mathrm{ref}}|}. \tag{61}$$

3. Experimental Methods and Results

The experiment was carried out using two related measurement configurations. The first configuration was used for controlled radio-frequency generation, frequency conversion, modulation, and reception of external radio signals. The second configuration was used for electron paramagnetic resonance measurements, in which the response of paramagnetic samples was recorded while an externally applied magnetic field was swept through the resonance region.

A RIGOL DG1022 two-channel function generator provided the controlled sinusoidal and modulated input signals. Frequency conversion was performed using an HP 10514A mixer. The input and output waveforms of the mixer were monitored in the time domain with a TiePie HS3 two-channel oscilloscope, while their frequency spectra were measured using an RSP1 software-defined radio receiver. The receiver was controlled through the HDSDR software, which displayed the measured signals as conventional spectra and as time-dependent waterfall diagrams.

Because the input of the software-defined radio receiver could be damaged by excessive signal power, a PRL T5075 variable attenuator was placed in the signal path when laboratory-generated signals were connected to the RSP1. The attenuation was adjusted before establishing the connection to the receiver and was reduced only when the signal level had been verified to be safe. The same receiver and software were subsequently used for measurements over the VLF, VHF, and UHF ranges.

An active VLF antenna was used for reception in the approximate frequency range from 10 kHz to 150 kHz. A separate laboratory VHF/FM antenna was used for reception of FM broadcast signals and selected transmissions in the VHF and UHF ranges. A PMR handheld radio and a wireless thermometer served as controlled local sources for selected narrow-band and burst-like transmissions.

For the EPR measurements, a modified low-noise block downconverter was used as the microwave source and receiving system. The sample was positioned inside the dielectric resonator associated with the LNB and placed between the pole pieces of an electromagnet. The microwave response of the resonator–sample system was converted to an intermediate-frequency signal near 800 MHz, which was received by the RSP1 and recorded using HDSDR.

The static magnetic field was produced by the electromagnet and controlled using a PS2000 programmable power supply. Predefined field sequences were applied in order to sweep the magnetic flux density through the expected resonance region. The progression of the resonance was observed as a characteristic curved trace in the HDSDR waterfall display. DPPH samples of different masses and ultramarine samples with different dilution ratios were inserted into the same resonator position as reproducibly as possible.

A second observation configuration employed a calibrated PLL-LNB with a specified local-oscillator frequency of 9.75 GHz. This configuration retained the same basic sample, magnet, receiver, and HDSDR arrangement, but provided a more accurately defined receiver local oscillator for reconstruction of the microwave frequency.

The main components and their functions are summarised in Table 1.

Table 1: Main components used for the RF-reception and EPR measurements.

Component	Function
RIGOL DG1022	Generation of two independent sinusoidal or modulated RF signals
HP 10514A mixer	Non-linear frequency conversion and generation of sum and difference frequencies
TiePie HS3	Two-channel recording of time-domain input and output voltages
RSP1 receiver	Digital reception and spectral acquisition over the investigated frequency ranges
HDSDR software	Receiver control, spectrum display, waterfall recording, tuning, bandwidth selection, and signal observation
PRL T5075 attenuator	Input attenuation and protection of the RSP1 receiver
Active VLF antenna	Reception of signals between approximately 10 kHz and 150 kHz
VHF/FM antenna	Reception of FM broadcast, VHF, PMR, and UHF signals
PMR radio and wireless thermometer	Controlled local transmission sources in the VHF/UHF ranges
Modified LNB and dielectric resonator	Microwave excitation, interaction with the paramagnetic sample, and downconversion of the EPR response
Calibrated PLL-LNB	EPR reception using a defined 9.75 GHz local oscillator
Electromagnet and PS2000 supply	Generation and programmed variation of the static magnetic field
DPPH samples	Reference samples and investigation of the dependence on sample mass
Ultramarine samples	Investigation of the EPR response at different dilution ratios

3.1 Frequency mixing and modulation

3.1.1 Measurement procedure

Channel 1 of the DG1022 was connected to input L of the HP 10514A mixer and set to a sinusoidal signal near 10 MHz. Channel 2 was connected to input R and set near 200 kHz. The mixer output X and the input signals were observed with the TiePie HS3. The output spectrum was then measured with the RSP1 and HDSDR. Before connecting the RSP1, the attenuator was set to at least 60 dB attenuation as required by the task sheet [**epr_tasks**].

For two sinusoidal input signals with frequencies f_1 and f_2, an ideal mixer produces frequency components at the sum and difference frequencies,

$$f_{\pm} = f_1 \pm f_2.$$

For the initial measurement with $f_1 = 10\,\text{MHz}$ and $f_2 = 200\,\text{kHz}$, the expected mixing products are therefore

$$f_- = 9.8\,\text{MHz} \qquad \text{and} \qquad f_+ = 10.2\,\text{MHz}.$$

The oscilloscope output for this configuration is shown in Fig. 6. The mixer output exhibits the expected beat-like waveform resulting from the superposition of the two frequency components.

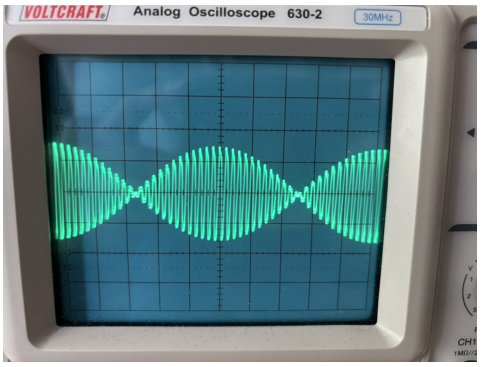


Figure 6: Oscilloscope output of the mixer for $f_1 = 10\,\text{MHz}$ and $f_2 = 200\,\text{kHz}$.

The corresponding frequency spectrum recorded with HDSDR is shown in Fig. 7. As expected, two dominant peaks appear near 9.8 MHz and 10.2 MHz, corresponding to the difference and sum frequencies, respectively.

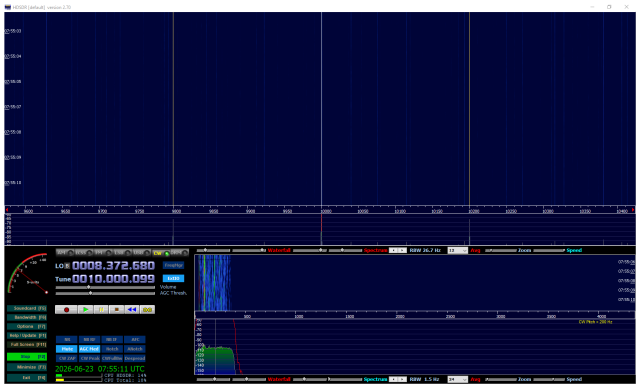


Figure 7: HDSDR spectrum for $f_1 = 10\,\text{MHz}$ and $f_2 = 200\,\text{kHz}$. The principal mixing products appear at $f_1 - f_2$ and $f_1 + f_2$.

To demonstrate the frequency conversion process, the second generator frequency f_2 was then varied while keeping $f_1 = 10\,\text{MHz}$ constant. The resulting HDSDR waterfall is shown in Fig. 8. As the value of f_2 increases, the two mixing products move symmetrically away from the carrier frequency according to

$$f_{\pm} = f_1 \pm f_2.$$

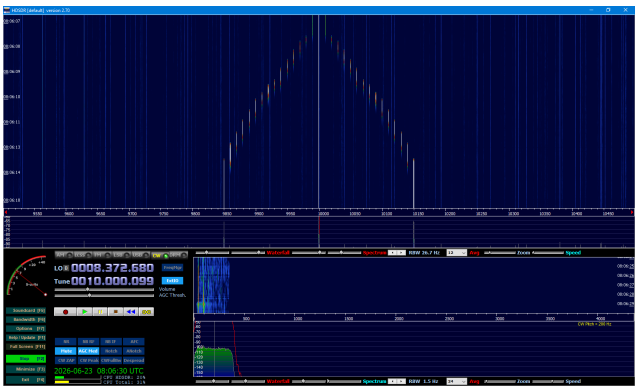


Figure 8: Waterfall spectrum obtained while varying f_2. The lower branch corresponds to $f_1 - f_2$, whereas the upper branch corresponds to $f_1 + f_2$.

Figure 8 clearly shows the two principal mixing branches. The lower-frequency branch corresponds to the difference frequency $(f_1 - f_2)$, while the upper branch corresponds to the sum frequency $(f_1 + f_2)$. Their symmetric separation about the carrier frequency confirms the expected operation of the frequency mixer. Besides these branches, a stationary component remains visible near 10 MHz. This signal is attributed to feedthrough of the original input signal through the non-ideal mixer, while additional weak spectral lines are likely caused by harmonics, intermodulation products, or external interference.

3.2 Receiving and analysing radio signals

3.2.1 Measurement procedure

A PMR walkie-talkie was used as the radio transmitter while the transmitted signal was received using the RSP1 receiver and analysed with the HDSDR software. The walkie-talkie was successively switched through its available channels while transmitting continuously. For each channel, the corresponding carrier frequency was recorded in the waterfall display.

3.2.2 Results

Figure 9 shows the waterfall spectrum recorded while the walkie-talkie channels were changed. Each bright vertical line corresponds to one transmission channel. As the channel number increases, the carrier frequency shifts systematically towards lower frequencies, producing the characteristic staircase pattern visible in the waterfall display.

A strong carrier common to all measurements remains visible at approximately 41.61 MHz, while the transmitted channel frequencies decrease from approximately 41.70 MHz to 41.69 MHz. The spacing between adjacent signals is nearly constant, indicating the fixed channel spacing implemented in the PMR radio. The waterfall display clearly demonstrates how changing the selected channel shifts the carrier frequency by discrete frequency steps.

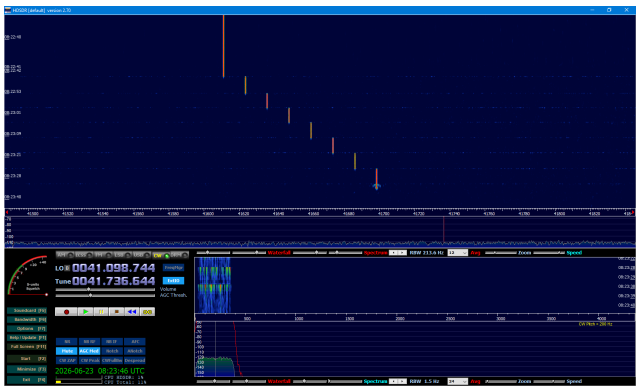


Figure 9: HDSDR waterfall spectrum recorded while switching through the channels of the PMR walkie-talkie. Each vertical line corresponds to one transmission channel.

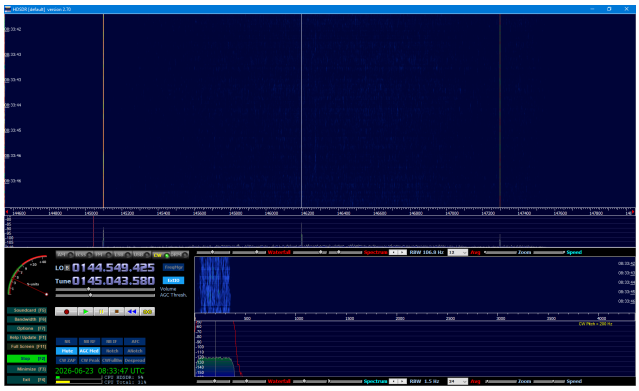


Figure 10: HDSDR spectrum recorded in the VHF band near 144.434 MHz. Several narrow-band signals characteristic of CW transmissions are visible.

3.3 FM, VHF and UHF reception

3.3.1 Measurement procedure

The laboratory VHF/FM antenna was connected to the RSP1 receiver through the PRL T5075 attenuator, which was set to at least 40 dB attenuation. The FM broadcast band was first explored using the maximum observation bandwidth of the receiver. Subsequently, the frequency region around 144.434 MHz was investigated for a continuous-wave (CW) signal, and additional measurements were carried out in the PMR and UHF frequency ranges.

3.3.2 Results

Figure 10 shows the spectrum recorded in the frequency range between approximately 144 MHz and 145 MHz. Several narrow spectral lines are clearly visible above the background noise. In contrast to FM broadcast signals, these carriers are very narrow in frequency and remain nearly constant over the recording time, which is characteristic of continuous-wave (CW) transmissions.

Although the task suggested searching for a CW signal at 144.434 MHz, no prominent transmission was observed exactly at this frequency. Instead, several nearby narrow-band carriers were detected within the observed frequency range.

Their small bandwidth compared with FM broadcast signals is consistent with CW operation.

During the measurement, the FM broadcast band was also explored. In contrast to the CW signals, FM stations occupied considerably wider bandwidths due to the frequency modulation used for audio transmission. Additional measurements in the PMR and UHF ranges demonstrated the presence of further radio services operating in these frequency bands.

3.4 Determination of the resonance frequency and Landé g-factor of DPPH

3.4.1 Measurement procedure

The DPPH sample was placed on the dielectric resonator of the modified low-noise block converter (LNB), while the external magnetic field was swept using the electromagnet. The intermediate-frequency (IF) output of the LNB was recorded with the RSP1 receiver and displayed in the HDSDR software as a waterfall spectrum.

Three independent measurements were recorded during repeated magnetic-field sweeps. Since the screenshots were not labelled during the experiment, they are referred to as Measurements 1–3 throughout this report.

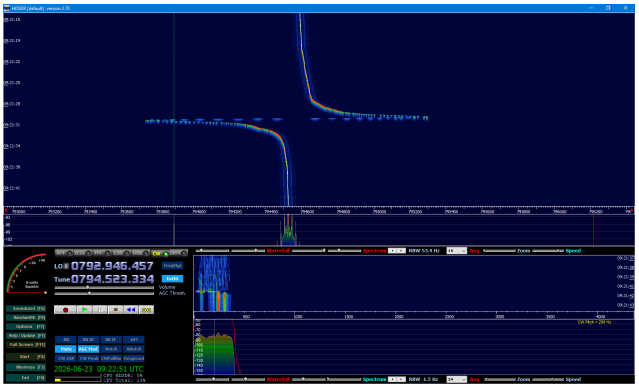


Figure 11: Raw HDSDR waterfall image for Measurement 1.

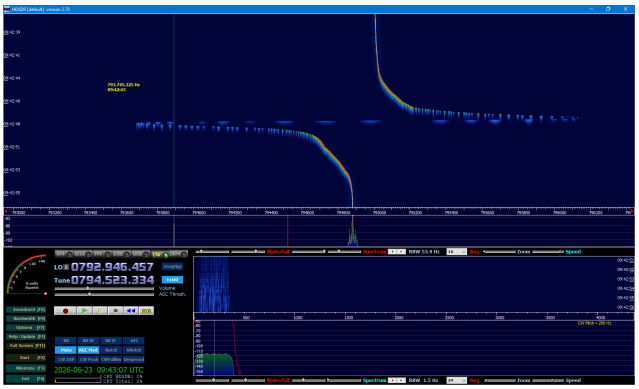


Figure 12: Raw HDSDR waterfall image for Measurement 2.

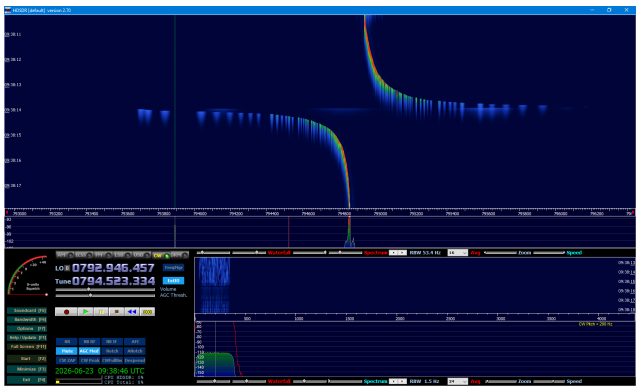


Figure 13: Raw HDSDR waterfall image for Measurement 3.

3.4.2 Data analysis

The centre of the dispersive resonance was extracted from each waterfall image using a Python image-processing routine. The horizontal pixel coordinate was converted into intermediate frequency using the calibrated HDSDR frequency axis. The vertical pixel coordinate was converted into magnetic field by assuming a linear magnetic-field sweep during the measurement and calibrating the start and end of the sweep from the recorded magnetometer values.

The extracted data were fitted using the dispersion model

$$f_{\mathrm{IF}}(B) = f_0 + \frac{A}{B - B_0}, \tag{62}$$

where f_0 is the horizontal asymptote, B_0 is the resonance magnetic field and A determines the curvature of the resonance.

The microwave resonance frequency was then calculated from

$$f_{\mathrm{res}} = f_{\mathrm{LO}} + f_{\mathrm{IF}}, \tag{63}$$

using the local oscillator frequency $f_{\mathrm{LO}} = 9.75\,\mathrm{GHz}$.

The fitted dispersion curves for all three measurements are shown below.

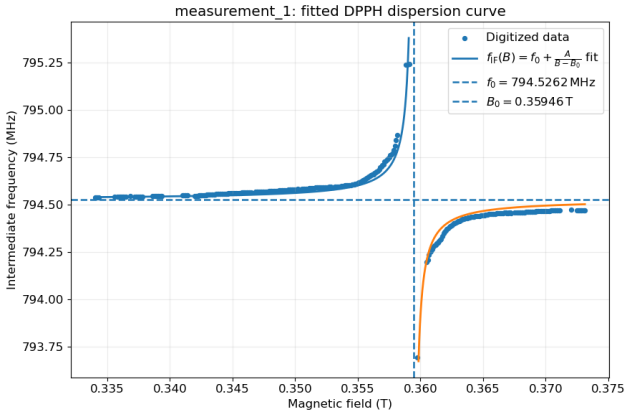


Figure 14: Dispersion fit for Measurement 1. The dashed lines indicate the fitted resonance field B_0 and the horizontal asymptote f_0.

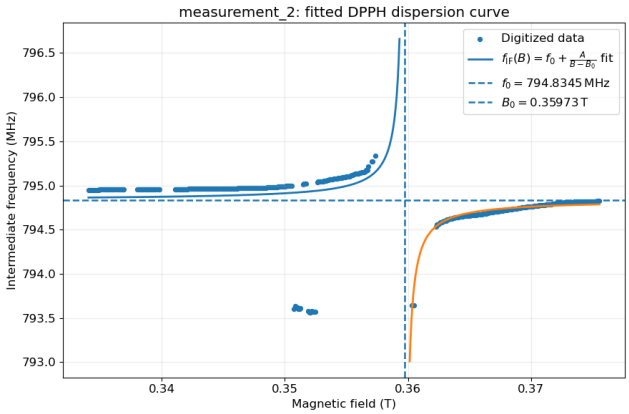


Figure 15: Dispersion fit for Measurement 2.

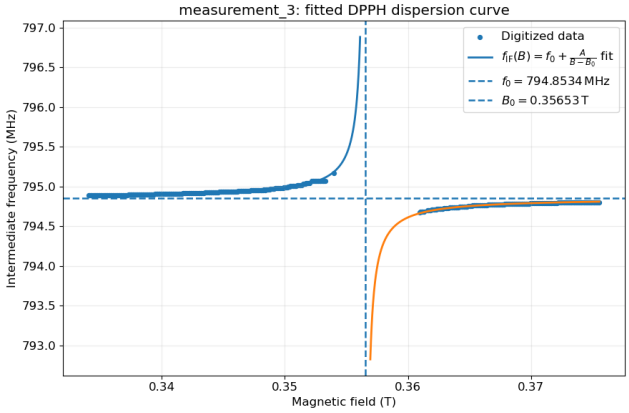


Figure 16: Dispersion fit for Measurement 3.

3.4.3 Results

The Landé factor was calculated from the EPR resonance condition

$$g = \frac{h f_{\mathrm{res}}}{\mu_B B_0}, \tag{64}$$

where h is Planck's constant and μ_B is the Bohr magneton. The uncertainties of the fitted parameters were obtained from the covariance matrix of the nonlinear least-squares fit, while the uncertainty of the g-factor was determined by standard error propagation.

The fitted resonance parameters for the three measurements are summarized in Table 2.

Table 2: Fitted resonance parameters and calculated Landé factors.

Measurement	B_0 (T)	g
1	0.35946 ± 0.00001	2.0959 ± 0.0001
2	0.35973 ± 0.00012	2.0944 ± 0.0007
3	0.35653 ± 0.00002	2.1132 ± 0.0001

The fitted resonance fields are consistent across all three measurements, lying between approximately 0.357 T and 0.360 T. Likewise, the resonance frequencies differ by less than 0.04 MHz, indicating good reproducibility of the fitted resonance position.

The calculated Landé factors are also mutually consistent, with values close to $g \approx 2.10$. The remaining variation between the measurements is primarily attributed to uncertainties in the magnetic-field calibration, the assumption of a linear field sweep during the measurement, and the finite resolution of the HDSDR screenshots used for extracting the dispersion curve.

3.5 Ultramarine dilution series

3.5.1 Measurement procedure

The EPR response of ultramarine was measured for the undiluted material and for dilution ratios of 1:10, 1:100, and 1:1000. Each sample was successively inserted into the dielectric resonator of the modified LNB and positioned between the pole pieces of the electromagnet. The sample position was reproduced as closely as possible in order to minimise changes in microwave coupling and resonator loading between measurements.

For each sample, the magnetic field was varied using the same programmed sequence of 81 increasing steps, with a duration of 1 s per step. The sequence covered an approximate magnetic-field range from 185 mT to 385 mT, followed by a return to a lower field value. The corresponding sweep rate was

$$\frac{\mathrm{d}B}{\mathrm{d}t} = \frac{385\,\mathrm{mT} - 185\,\mathrm{mT}}{81\,\mathrm{s}} = 2.469\,\mathrm{mT/s}. \tag{65}$$

The microwave response was received using the RSP1 software-defined radio and recorded in HDSDR as a waterfall diagram. The horizontal coordinate represented the down-converted intermediate frequency, while the vertical coordinate represented acquisition time. The receiver settings were kept as similar as practically possible, although the HDSDR centre frequency was adjusted between some measurements in order to keep the dispersion trace within the visible frequency window.

The bright central ridge of each waterfall image was digitised to obtain the intermediate frequency f_{IF} as a function of time. The waterfall time coordinate was converted to magnetic flux density using the linear field-sweep relation introduced in Eq. (34), with $t = 0$ assigned to 185 mT and $t = 81$ s assigned to 385 mT.

For recordings in which the end-of-sequence jump was visible, the jump was used to align the waterfall time with the final field step. For weaker traces without an equally distinct reset feature, the final visible part of the recorded sweep was associated with the end of the programmed sequence. This alignment was included in the uncertainty assigned to the reconstructed field coordinate.

A cubic smoothing spline was applied to each digitised trace in order to reduce small pixel-scale fluctuations and to

identify the characteristic branches of the measured feature. The spline was used only as an empirical guide to the centre of the dispersion curve and was not treated as a theoretical EPR line-shape function.

For the pure and 1:10 samples, both sides of the measured feature were sufficiently well defined to determine the centre from the midpoint between two characteristic points,

$$B_0 = \frac{B_- + B_+}{2}, \qquad f_{\mathrm{IF},0} = \frac{f_- + f_+}{2}. \tag{66}$$

Here, B_- and B_+ denote the magnetic-field coordinates of the selected lower and upper branches, while f_- and f_+ are the corresponding intermediate frequencies.

The 1:100 and 1:1000 measurements produced weaker and more step-like traces. For these recordings, the central intermediate frequency was estimated from the midpoint between the low-frequency and high-frequency branches,

$$f_{\mathrm{IF},0} = \frac{f_{\mathrm{low}} + f_{\mathrm{high}}}{2}. \tag{67}$$

The corresponding magnetic-field coordinate was determined from the crossing condition

$$f_{\mathrm{spline}}(B_0) = f_{\mathrm{IF},0}. \tag{68}$$

The experimental g-factor was calculated using Eq. (31). The nominal microwave excitation frequency of the direct-mixing system was taken to be

$$\nu_{\mathrm{mw}} = 10.6\,\mathrm{GHz}. \tag{69}$$

The percentage deviation from the adopted reference value $g_{\mathrm{ref}} = 2.0290$ was calculated using Eq. (61).

The uncertainties assigned to the extracted centre fields included the estimated sensitivity to ridge digitisation, centre-point selection, smoothing, and time-to-field alignment. Possible common systematic uncertainties in the absolute magnetic-field scale and in the nominal microwave frequency were not included in the individual uncertainties.

3.5.2 Results

A field-dependent dispersion feature was detected for all four ultramarine samples. The complete digitised traces and their corresponding smoothing splines are shown in Fig. 17.

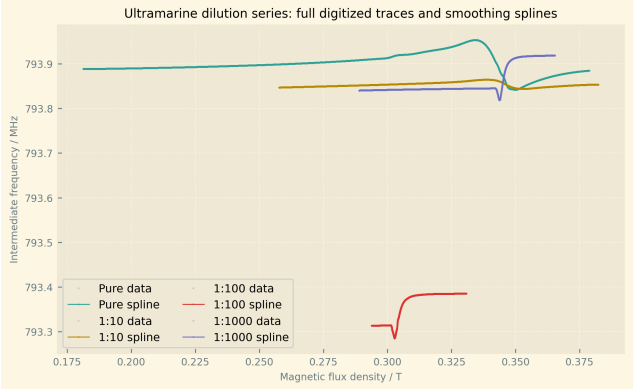
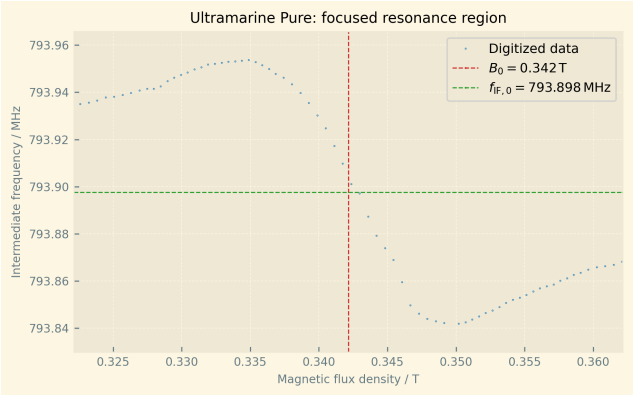
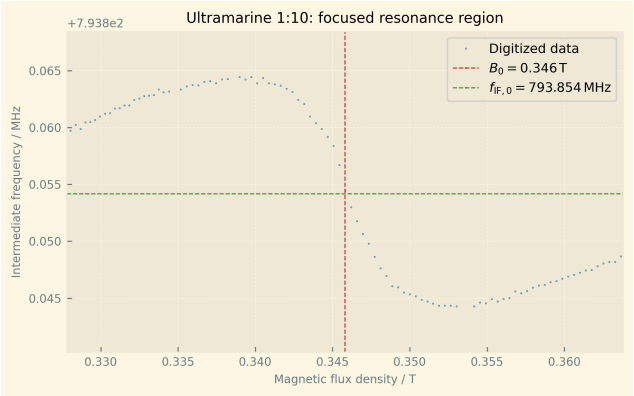


Figure 17: Complete digitised ultramarine dispersion traces for the pure, 1:10, 1:100, and 1:1000 samples. The continuous curves are cubic smoothing splines used to identify the characteristic branches and transition regions. The different intermediate-frequency baselines result partly from changes in the HDSDR tuning between recordings.

The focused regions used to determine the centre coordinates are shown in Fig. 19. The digitised data are represented by blue points, the extracted centre field B_0 is indicated by a red vertical line, and the intermediate-frequency centre $f_{\mathrm{IF},0}$ is indicated by a green horizontal line.

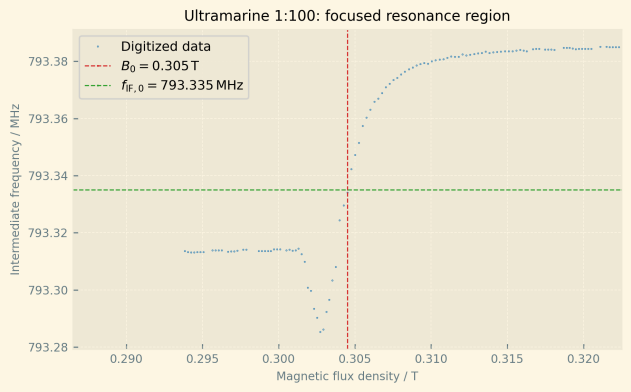


(a) Pure ultramarine.

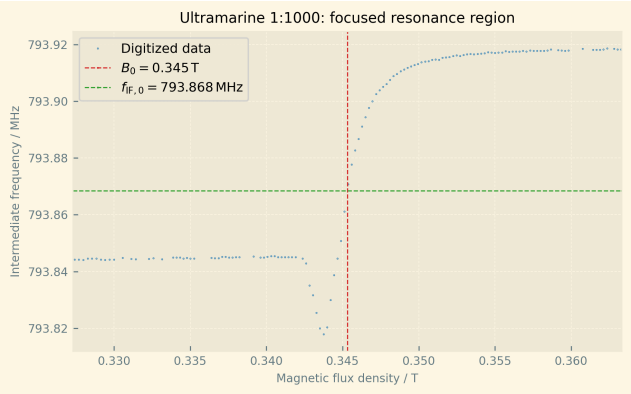


(b) 1:10 dilution.

Figure 18: Focused regions used to extract the centre coordinates of the ultramarine dispersion features. The pure and 1:10 samples were evaluated using Eq. (66).



(a) 1:100 dilution.



(b) 1:1000 dilution.

Figure 19: Focused regions used to extract the centre coordinates of the ultramarine dispersion features. The 1:100 and 1:1000 samples were evaluated using Eq. (66).

The extracted intermediate-frequency centres, reconstructed magnetic-field coordinates, and calculated g-factors are summarised in Table 3.

Table 3: Extracted resonance parameters for the ultramarine dilution series. The field coordinates were reconstructed using the assumed 81-step linear magnetic-field sweep.

Sample	B_0 / T	$f_{\mathrm{IF},0}$ / MHz	g_{exp}	Dev. / %
Pure	0.342 ± 0.006	793.898 ± 0.001	2.214 ± 0.039	9.1
1:10	0.346 ± 0.006	793.854 ± 0.001	2.190 ± 0.038	7.9
1:100	0.305 ± 0.012	793.335 ± 0.002	2.487 ± 0.098	22.6
1:1000	0.345 ± 0.012	793.868 ± 0.002	2.193 ± 0.076	8.1

The reconstructed centre fields of the pure, 1:10, and 1:1000 samples lay between approximately 0.342 T and 0.346 T. Their calculated g-factors ranged from approximately 2.19 to 2.21.

The 1:100 measurement yielded a reconstructed centre field of approximately 0.305 T and a calculated g-factor of approximately 2.49. This result differed substantially from the other three measurements and is examined separately in Section **??**.

The experimental g-factors are compared with the adopted reference value in Fig. 20.

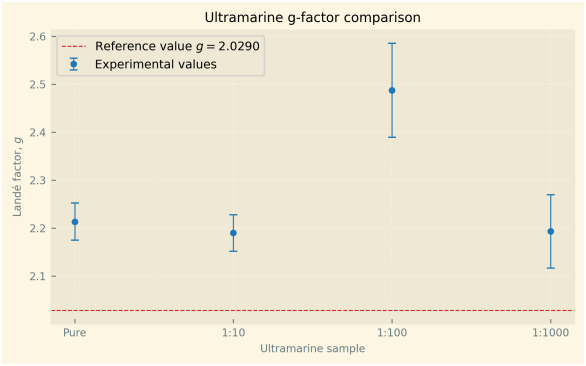


Figure 20: Calculated ultramarine g-factors compared with the adopted reference value $g_{\mathrm{ref}} = 2.0290$.

No calibrated signal amplitude was extracted from the HDSDR colour scale because the displayed intensity depended on receiver gain, waterfall scaling, background level, and resonator coupling. The results were therefore restricted to the digitised frequency trajectory and its reconstructed centre coordinates.

An independent linewidth was likewise not determined from the waterfall traces. The measured curves represented the response of the complete modified-LNB, resonator, and receiver system during the field sweep rather than a directly calibrated Lorentzian absorption spectrum.

3.6 Comparison of direct-mixing and calibrated PLL-LNB detection

3.6.1 Measurement procedure

The DPPH sample was measured using the calibrated PLL-LNB in order to compare the reconstructed microwave frequency with the direct-mixing result obtained in Section 3.4. The local-oscillator frequency of the calibrated PLL-LNB was specified as

$$f_{\mathrm{LO,PLL}} = 9.75\,\mathrm{GHz}. \tag{70}$$

The sample was placed on the dielectric resonator between the pole pieces of the electromagnet. The magnetic field was swept through the DPPH resonance region while the down-converted microwave response was recorded using the RSP1 receiver and HDSDR.

Two usable calibrated PLL-LNB waterfall recordings were available. The horizontal coordinate of each screenshot was converted into intermediate frequency using the labelled HDSDR frequency ticks, while the vertical coordinate was converted into elapsed time using the displayed timestamps.

The bright central ridge of each asymptotic trajectory was digitised and fitted with the empirical reciprocal relation

$$f_{\mathrm{IF}}(t) = f_{\mathrm{IF},0} + \frac{C}{t - t_0}, \tag{71}$$

where $f_{\mathrm{IF},0}$ is the horizontal intermediate-frequency asymptote, t_0 is the fitted resonance-time asymptote, and C is an empirical curvature parameter.

Equation (71) was used only to determine the asymptotic coordinates of the measured waterfall trajectory. It was not interpreted as an intrinsic EPR absorption, dispersion, or derivative line shape.

For comparison with the direct-mixing arrangement, measurement 2 from Section 3.4 was used. Its fitted intermediate-frequency asymptote and resonance field were

$$f^{(2)}_{\mathrm{IF,direct}} = (794.834548 \pm 0.014117)\,\mathrm{MHz}, \tag{72}$$

and

$$B^{(2)}_{0,\mathrm{direct}} = (0.359732 \pm 0.000118)\,\mathrm{T}. \tag{73}$$

The fitted uncertainty in Eq. (73) represents only the statistical uncertainty of the hyperbolic fit. It does not include the systematic uncertainty of the reconstructed magnetic-field scale.

The microwave frequency in each observation system was reconstructed using the upper mixing branch,

$$f_{\mathrm{RF}} = f_{\mathrm{LO}} + f_{\mathrm{IF}}. \tag{74}$$

For the calibrated PLL-LNB, the specified local-oscillator frequency in Eq. (70) was used. For the direct-mixing arrangement, the same nominal value of 9.75 GHz was used, although this oscillator was not independently calibrated.

Consequently, the uncertainty quoted for the direct-mixing microwave frequency includes the fitted intermediate-frequency uncertainty but does not include an unknown systematic offset or drift of the direct-mixing local oscillator.

3.6.2 Results

Both calibrated PLL-LNB recordings displayed clearly resolved asymptotic DPPH trajectories. The digitised data and reciprocal fits are shown in Figure 21.

The fitted intermediate-frequency asymptotes were

$$f_{\mathrm{IF,PLL},1} = 793.514\,337\,\mathrm{MHz}, \tag{75}$$

and

$$f_{\mathrm{IF,PLL},2} = 793.369\,368\,\mathrm{MHz}. \tag{76}$$

The mean intermediate frequency and the sample standard deviation of the two recordings were therefore

$$\overline{f}_{\mathrm{IF,PLL}} = (793.441853 \pm 0.102509)\,\mathrm{MHz}. \tag{77}$$

Using Eq. (74), the calibrated PLL-LNB microwave frequency was

$$\begin{aligned} f_{\mathrm{RF,PLL}} &= 9.75\,\mathrm{GHz} + \overline{f}_{\mathrm{IF,PLL}} \\ &= (10.543441853 \pm 0.000102509)\,\mathrm{GHz}. \end{aligned} \tag{78}$$

For direct-mixing measurement 2, the reconstructed microwave frequency was

$$\begin{aligned} f^{(2)}_{\mathrm{RF,direct}} &= 9.75\,\mathrm{GHz} + f^{(2)}_{\mathrm{IF,direct}} \\ &= (10.544834548 \pm 0.000014117)\,\mathrm{GHz}. \end{aligned} \tag{79}$$

The results are summarised in Table 4.

The nominal absolute frequency difference between the two observation systems was

$$\begin{aligned} \Delta f_{\mathrm{RF}} &= \left| f^{(2)}_{\mathrm{RF,direct}} - f_{\mathrm{RF,PLL}} \right| \\ &= (1.392696 \pm 0.103476)\,\mathrm{MHz}. \end{aligned} \tag{80}$$

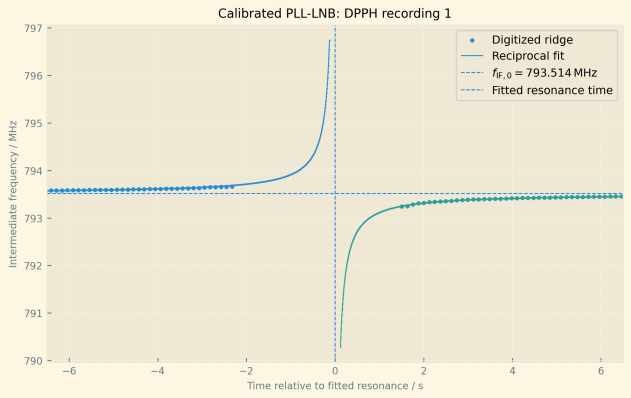
The uncertainty in Eq. (80) was obtained by combining the fitted uncertainty of direct-mixing measurement 2 and the repeatability uncertainty of the PLL-LNB measurements in quadrature.

The corresponding relative frequency difference, evaluated with respect to the calibrated PLL-LNB frequency, was

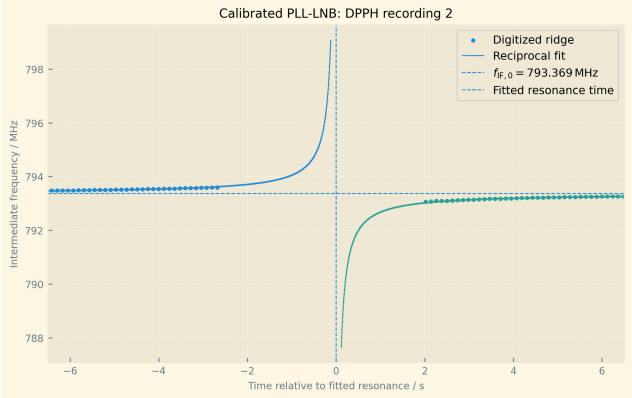
$$\begin{aligned} \delta f &= \frac{\left| f^{(2)}_{\mathrm{RF,direct}} - f_{\mathrm{RF,PLL}} \right|}{f_{\mathrm{RF,PLL}}} \times 100 \\ &= 0.013\,209\,\%. \end{aligned} \tag{81}$$

The reconstructed frequencies therefore differed by only approximately 0.013 %. However, the significance of this difference cannot be evaluated solely from the quoted statistical uncertainties because the absolute frequency offset and drift of the direct-mixing local oscillator were not independently known.

For comparison of the calculated g-factors, the resonance field of direct-mixing measurement 2 was first combined



(a) Calibrated PLL-LNB recording 1.



(b) Calibrated PLL-LNB recording 2.

Figure 21: Digitised DPPH waterfall trajectories measured using the calibrated PLL-LNB, shown symmetrically around the fitted resonance time. The continuous curves represent the reciprocal fit in Eq. (71). The vertical dashed lines indicate the fitted resonance times, while the horizontal dashed lines indicate the intermediate-frequency asymptotes.

Table 4: Comparison of direct-mixing measurement 2 with the calibrated PLL-LNB DPPH measurement. The quoted frequency uncertainties include fit repeatability but not the unknown systematic offset of the direct-mixing local oscillator.

Observation system	$f_{\mathrm{IF},0}$ / MHz	f_{RF} / GHz	B_0 / T	g
Direct mixing, measurement 2	794.834548 ± 0.014117	10.544834548 ± 0.000014117	0.359732 ± 0.000118	2.094349
Calibrated PLL-LNB	793.441853 ± 0.102509	10.543441853 ± 0.000102509	0.359732 ± 0.000118[a]	2.094073[a]

[a] The Task 4 measurement 2 field was used only to evaluate the effect of the changed microwave frequency. It was not measured independently during the PLL-LNB recording.

with its own reconstructed microwave frequency, giving

$$g^{(2)}_{\mathrm{direct}} = \frac{h f^{(2)}_{\mathrm{RF,direct}}}{\mu_{\mathrm{B}} B^{(2)}_{0,\mathrm{direct}}} = 2.094349. \tag{82}$$

Using the same field coordinate together with the calibrated PLL-LNB frequency gives

$$g_{\mathrm{PLL,same\,field}} = \frac{h f_{\mathrm{RF,PLL}}}{\mu_{\mathrm{B}} B^{(2)}_{0,\mathrm{direct}}} = 2.094073. \tag{83}$$

The small difference between Eqs. (82) and (83) shows that replacing the nominal direct-mixing frequency by the calibrated PLL-LNB frequency changes the calculated g-factor only slightly. The much larger deviation from the DPPH literature value must therefore arise mainly from the reconstructed magnetic-field coordinate rather than from the approximately 1.39 MHz frequency difference.

Using the calibrated PLL-LNB frequency and the adopted DPPH reference value

$$g_{\mathrm{DPPH}} = 2.0036, \tag{84}$$

the expected resonance field was

$$B_{\mathrm{res,expected}} = \frac{h f_{\mathrm{RF,PLL}}}{g_{\mathrm{DPPH}} \mu_{\mathrm{B}}} = 375.976\,\mathrm{mT}. \tag{85}$$

The fitted resonance field of direct-mixing measurement 2 was

$$B^{(2)}_{0,\mathrm{direct}} = 359.732\,\mathrm{mT}. \tag{86}$$

The difference between the reconstructed Task 4 field and the field expected from the calibrated PLL-LNB frequency was therefore

$$\Delta B = B^{(2)}_{0,\mathrm{direct}} - B_{\mathrm{res,expected}} = -16.244\,\mathrm{mT}. \tag{87}$$

The corresponding relative deviation was

$$\varepsilon_B = \frac{\left| B^{(2)}_{0,\mathrm{direct}} - B_{\mathrm{res,expected}} \right|}{B_{\mathrm{res,expected}}} \times 100 = 4.320\,\%. \tag{88}$$

This field difference is considerably larger than the relative microwave-frequency difference. It therefore supports

the conclusion that the dominant contribution to the high experimental g-factor in measurement 2 is the absolute magnetic-field calibration rather than the frequency reconstruction.

The field comparison should nevertheless be regarded as an estimate. The field axis of measurement 2 was reconstructed from approximate start and end values obtained from the magnetometer video, assuming a linear sweep. The quoted fit uncertainty of B_0 does not include systematic effects from the field-scale endpoints, sweep non-linearity, magnet response delay, or imperfect synchronisation between the video and the HDSDR waterfall.

The correspondence between measurement 2 and the nominal DPPH sample masses was not preserved unambiguously. Measurement 2 was used here as requested, but the comparison should not be described as a confirmed 10 µg-to-10 µg comparison unless the sample assignment can be verified from the laboratory notes.

An intrinsic linewidth was not extracted because the waterfall trajectory represents the response of the complete resonator, LNB, mixer, and software-defined radio system during a magnetic-field sweep rather than a directly calibrated Lorentzian EPR absorption spectrum.

3.7 Data extraction and uncertainty

The measured data were available primarily as oscilloscope images, HDSDR spectra, and HDSDR waterfall screenshots. Frequencies, bandwidths, time coordinates, and resonance curves were therefore extracted either directly from the numerical values displayed by the instruments or by digitising the corresponding images. Where possible, repeated recordings were evaluated independently in order to estimate the repeatability of the complete measurement and analysis procedure.

For the RF measurements, spectral peak positions were read from the HDSDR frequency scale. The centre frequency of an isolated spectral line was taken from the position of its intensity maximum, while the occupied bandwidth was determined from the visible lower and upper frequency limits using Eq. (22). The uncertainty was estimated from the displayed frequency resolution, the finite line width, and the precision with which the spectral boundaries could be identified.

The time-domain voltage amplitudes were read from the oscilloscope scale. Peak-to-peak voltages were converted to root-mean-square voltages using Eq. (4), and the corresponding powers were calculated using Eq. (5). The principal uncertainties originated from the finite oscilloscope grid resolution, trace thickness, generator-amplitude accuracy, and the assumption of a correctly matched load impedance.

For the EPR measurements, the HDSDR waterfall screenshots were cropped and digitised by following the bright central ridge of the measured dispersion feature. The horizontal image coordinate was converted to intermediate frequency by selecting two known tick positions (x_1, f_1) and (x_2, f_2) from the displayed frequency axis. Assuming a linear scale, the frequency corresponding to a pixel coordinate x was calculated as

$$f_{\mathrm{IF}}(x) = f_1 + \frac{x - x_1}{x_2 - x_1}\,(f_2 - f_1)\,. \tag{89}$$

The local uncertainty introduced by the image coordinate was estimated from

$$u_{\mathrm{pix}}(f_{\mathrm{IF}}) = \left|\frac{f_2 - f_1}{x_2 - x_1}\right| u(x), \tag{90}$$

where $u(x)$ represents the estimated uncertainty in locating the centre of the digitised ridge. This uncertainty included the finite pixel size, the visible thickness of the trace, colour-threshold selection, and small fluctuations of the waterfall ridge.

The waterfall time coordinate was calibrated in an analogous way from the displayed timestamps. For two reference positions (y_1, t_1) and (y_2, t_2), the time corresponding to a vertical pixel coordinate y was calculated from

$$t(y) = t_1 + \frac{y - y_1}{y_2 - y_1}\,(t_2 - t_1)\,. \tag{91}$$

Where an unambiguous sequence start or reset was visible, the waterfall time was converted to magnetic field using the linear field-sweep relation in Eq. (34). The conversion assumed 81 one-second field steps between approximately 185 mT and 385 mT. The resulting nominal sweep rate was given by Eq. (65).

The field conversion introduced several uncertainty contributions. These included the approximate values of the initial and final magnetic fields, possible non-linearity of the electromagnet response, uncertainty in identifying the beginning and end of the sequence, the finite duration of each field step, and any delay between the programmed field change and the actual magnetic-field response. A further uncertainty arose because the magnetic field at the sample position was not recorded simultaneously in every HDSDR screenshot.

For recordings in which the absolute sequence alignment could not be established reliably, the measured data were retained as intermediate frequency versus time relative to the fitted resonance. In these cases, the intermediate-frequency asymptote remained measurable, but an independent absolute resonance field was not assigned.

The DPPH curves were evaluated using the reciprocal model in Eq. (71). The numerical fit returned the horizontal intermediate-frequency asymptote, the resonance-time asymptote, and their statistical covariance. The uncertainty of the final fitted value was not taken solely from the numerical fit covariance because this would underestimate the effects of image calibration, ridge selection, and run-to-run variation.

For the ultramarine samples, the centre coordinates were extracted either from the midpoint construction in Eq. (66) or from the spline-crossing method in Eqs. (67) and (68). The associated uncertainty was estimated by varying the selected branch points, smoothing strength, and field-sequence alignment within reasonable limits and recalculating the centre coordinates.

When two repeated recordings were available, the observed repeatability was characterised by the sample standard deviation

$$s(x) = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \overline{x})^2}{n-1}}, \tag{92}$$

where x_i denotes an individual result, $\overline{x}$ is the arithmetic mean, and n is the number of recordings. For the direct-mixing and PLL-LNB comparison, the quoted intermediate-frequency uncertainties were obtained from the two available recordings using Eq. (92).

The reconstructed microwave-frequency uncertainty was calculated using Eq. (56). For the calibrated PLL-LNB, the local-oscillator contribution was negligible compared with the spread of the measured intermediate frequencies. For the direct-mixing arrangement, the use of a nominal local-oscillator frequency introduced an additional systematic uncertainty that could not be determined from the screenshots alone.

The uncertainty of the experimental g-factor was evaluated using Eq. (55). The magnetic-field uncertainty generally contributed more strongly than the frequency uncertainty because the HDSDR intermediate frequency could be read with comparatively high precision, whereas the absolute field coordinate depended on the reconstructed field sequence and the magnetic-field calibration.

The uncertainty of differences between independently obtained values was calculated using Eq. (58). In general, independent statistical contributions were combined in quadrature,

$$u_{\mathrm{c}}(q) = \sqrt{\sum_i \left(\frac{\partial q}{\partial x_i} u(x_i) \right)^2}, \tag{93}$$

where q is the calculated quantity and x_i are the independent input quantities.

Systematic effects that could not be quantified from the recorded data were reported separately rather than being included as unrealistically precise numerical uncertainties. These included an unknown offset of the field display, oscillator drift in the direct-mixing arrangement, receiver retuning, uncertainty in the selected mixing sideband, sample positioning, resonator loading, and magnetic-field inhomogeneity.

An additional experimental limitation arose from the relative orientation of the static magnetic field B_0 and the microwave magnetic field B_1. Only the component of B_1 perpendicular to B_0 can efficiently drive the EPR transition. If θ is the angle between the two fields, the effective transverse component is

$$B_{1,\perp} = B_1 \sin\theta. \tag{94}$$

The ideal arrangement corresponds to $\theta = 90^\circ$. For an angular misalignment $\varepsilon = 90^\circ - \theta$,

$$\frac{B_{1,\perp}}{B_1} = \cos\varepsilon \approx 1 - \frac{\varepsilon^2}{2}, \tag{95}$$

where the approximation applies for small ε expressed in radians.

Because the angle between B_0 and B_1 was not measured directly, this effect could not be assigned a numerical uncertainty. Imperfect perpendicularity primarily reduces the microwave excitation efficiency and therefore decreases the signal amplitude and signal-to-noise ratio. It does not alter the fundamental resonance condition in Eq. (30) to first order, but a weaker signal increases the uncertainty with which the dispersion centre can be digitised and fitted.

Repeated insertion of the samples introduced additional uncertainty because small changes in sample position and orientation modify the microwave field experienced by the sample, the filling factor of the resonator, and the local static magnetic field. These effects were especially important for the strongly diluted ultramarine samples, whose dispersion traces were already close to the background fluctuations.

The HDSDR colour scale was not calibrated as an absolute signal-amplitude scale. The displayed intensity depended on receiver gain, attenuation, waterfall contrast, noise-floor settings, and resonator coupling. Consequently, the colour brightness of different screenshots was not used for a quantitative amplitude comparison.

An intrinsic EPR linewidth was also not extracted from the waterfall curves. The measured traces described the frequency response of the complete resonator, LNB, mixer, and software-defined radio chain during a magnetic-field sweep, rather than a directly calibrated absorption or first-derivative EPR spectrum. Any apparent width therefore contained both sample-dependent and instrumental contributions.

The principal uncertainty sources and their treatment are summarised in Table 5.

Table 5: Principal uncertainty sources in the RF and EPR measurements.

Source	Affected quantity	Treatment
HDSDR pixel calibration	f_{IF}, bandwidth, curve position	Estimated from pixel scale, ridge thickness, and tick interpolation.
Waterfall-time calibration	Resonance time and field coordinate	Estimated from timestamp spacing and sequence alignment.
Field-sequence endpoints	B_{res} and g	Treated as a systematic field-scale uncertainty.
Field non-linearity and response delay	Reconstructed field axis	Not independently calibrated; treated as a systematic limitation.
Local-oscillator accuracy and drift	Microwave frequency	Calibrated for the PLL-LNB; nominal and unquantified for direct mixing.
Repeated recordings	Frequency and fit repeatability	Characterised using the sample standard deviation.

Table 5: Principal uncertainty sources in the RF and EPR measurements, continued.

Source	Affected quantity	Treatment
Ridge selection and fit model	Asymptotes and centre coordinates	Assessed through fit residuals and variation of the extraction choices.
Sample position and resonator loading	Signal amplitude, centre stability, and SNR	Reduced through reproducible placement; remaining contribution treated systematically.
Non-perpendicular B_0 and B_1	Excitation efficiency and SNR	Not quantified because the misalignment angle was not measured.
HDSDR gain and colour scale	Apparent signal amplitude	No absolute amplitude comparison was performed.
Magnetic-field inhomogeneity	Resonance position and apparent width	Included as an unquantified systematic contribution.

4. Discussion

The experiment combined two complementary topics of applied physics. The first part investigated the principles of radio-frequency signal generation, frequency conversion, modulation and software-defined radio reception, while the second part applied these concepts to continuous-wave electron paramagnetic resonance spectroscopy. Together, the measurements demonstrated how high-frequency electromagnetic signals can be generated, detected, converted to lower frequencies and finally used to probe the magnetic properties of paramagnetic materials.

In Task 1, the operation of the frequency mixer agreed well with the theoretical expectation. The oscilloscope showed the characteristic beat-like waveform produced by mixing two sinusoidal signals, while the HDSDR spectrum contained the expected sum and difference frequencies. The symmetric movement of the two mixing products as the second generator frequency was varied confirmed the relation

$$f_{\pm} = f_1 \pm f_2.$$

Besides the desired mixing products, weak leakage of the original carrier and additional spectral components were observed, illustrating the non-ideal behaviour of practical mixers due to finite isolation, harmonic generation and intermodulation.

The radio reception measurements demonstrated the capabilities of a software-defined radio for analysing signals over a wide frequency range. Stable narrow-band carriers characteristic of continuous-wave transmissions were observed in the VHF band, whereas FM broadcast stations occupied significantly larger bandwidths because of frequency modulation. The waterfall displays also illustrated how different modulation schemes can be distinguished from their temporal behaviour. These observations demonstrate one of the principal advantages of SDR systems, namely that signal processing, filtering and demodulation can be performed entirely in software while preserving the complete received spectrum.

The DPPH measurements produced reproducible dispersion curves from which the resonance field and microwave frequency were extracted. All three measurements yielded resonance fields near 0.36 T and calculated Landé factors close to $g \approx 2.10$. Although the measurements were internally consistent, the obtained values are larger than the literature value of $g_{\mathrm{DPPH}} \approx 2.0036$. Comparison with the calibrated PLL-LNB measurements showed that the reconstructed microwave frequencies differed by only about 0.013 %, whereas the reconstructed resonance field differed by approximately 16 mT from the expected value. This indicates that the dominant systematic uncertainty originated from the magnetic-field calibration and the reconstruction of the field sweep rather than from the microwave-frequency determination. The assumptions of a perfectly linear field sweep, approximate start and end field values, and image-based digitisation of the HDSDR waterfall therefore constitute the largest uncertainty sources.

The ultramarine dilution series demonstrated that EPR signals remained detectable even after substantial dilution of the sample. The pure, 1:10 and 1:1000 samples produced similar resonance positions, indicating that dilution itself does not significantly change the electronic environment of the paramagnetic centres. In contrast, the 1:100 sample yielded a noticeably different g-factor, which is most likely explained by the comparatively weak signal, uncertainties in determining the centre of the dispersion feature and limitations of the field reconstruction. Since the HDSDR colour scale is not calibrated to absolute signal intensity, only qualitative conclusions regarding the reduction of signal strength with dilution could be drawn.

The comparison between the direct-mixing arrangement and the calibrated PLL-LNB highlighted the importance of a stable local oscillator for quantitative spectroscopy. The calibrated PLL-LNB provided a well-defined microwave frequency and excellent frequency stability. However, replacing the direct-mixing frequency with the calibrated frequency changed the calculated g-factor only slightly. This confirms that improving the frequency calibration alone is insufficient to obtain accurate g-factors unless the magnetic-field calibration is also known with comparable accuracy. Future improvements would therefore benefit primarily from an independently calibrated magnetic-field

measurement using, for example, a Hall probe or NMR gaussmeter during the field sweep.

Overall, the experiment demonstrated that meaningful spectroscopic information can be extracted even from relatively simple microwave hardware by combining software-defined radio techniques with numerical data analysis. At the same time, it illustrates that systematic uncertainties, particularly in the magnetic-field calibration, can dominate the final accuracy even when statistical fitting uncertainties are very small.

5. Conclusion

In this experiment, the principles of radio-frequency signal processing and electron paramagnetic resonance spectroscopy were investigated using software-defined radio techniques. Frequency mixing was successfully demonstrated through the observation of the expected sum and difference frequencies, confirming the operation of the nonlinear mixer. Radio reception measurements illustrated the different spectral characteristics of FM, continuous-wave and digital radio signals and demonstrated the flexibility of software-defined radio for analysing signals over a broad frequency range.

In the EPR part of the experiment, resonance curves of DPPH and ultramarine samples were successfully recorded using a modified LNB receiver. The resonance positions were extracted from HDSDR waterfall images using image processing and numerical fitting. Repeated DPPH measurements yielded reproducible resonance fields and Landé factors near $g \approx 2.10$, while the ultramarine dilution series showed that the EPR signal remained observable over a wide range of concentrations, although weaker signals resulted in larger uncertainties in the extracted resonance parameters.

Comparison of the direct-mixing system with the calibrated PLL-LNB demonstrated that the reconstructed microwave frequencies agreed to within approximately 0.013 %. The remaining deviation of the measured DPPH g-factor from the literature value was therefore attributed mainly to uncertainties in the magnetic-field calibration rather than to the microwave-frequency reconstruction. This emphasises that accurate EPR measurements require both a stable microwave source and a precisely calibrated magnetic-field scale.

Overall, the experiment provided practical insight into both modern radio-frequency receiver technology and continuous-wave EPR spectroscopy. It demonstrated how heterodyne detection, software-defined radio and numerical signal analysis can be combined to investigate the magnetic properties of paramagnetic materials. The laboratory also highlighted the importance of systematic uncertainty analysis and showed that, in spectroscopic experiments, the overall measurement accuracy is often limited by calibration procedures rather than by statistical precision alone.

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